

Data Analysis with Spreadsheets and Python

TEC-004 · Week 1 of 10 · IT and Technology

Study independently · keep a notebook · save your evidence



THIS WEEK'S OUTCOME

Prepare a clean analysis dataset.

STUDY RHYTHM

6 hours

Short sessions work best: Read · Watch · Make · Reflect.

START HERE · IN PLAIN LANGUAGE

In plain language, this lesson helps you work on prepare a clean analysis dataset.. Treat foundations and workflow as a skill you can build in small steps: study one idea, compare it with a real Liberian or regional example, and save a short piece of evidence of what you understood.

01

READ

Find the main idea.

02

WATCH

Notice a method.

03

MAKE

Try it locally.

04

REFLECT

Check what changed.

Your learning path

READ FOR THE IDEA

Read for the main idea: Prepare a clean analysis dataset.

WATCH FOR THE METHOD

Watch for one method or example connected to foundations and workflow.

MAKE EVIDENCE

Apply the idea locally and save analysis notebook and dashboard.

ESSENTIAL QUESTION

How could foundations and workflow improve a real situation connected to Data Analysis with Spreadsheets and Python?

Write one sentence before you begin. Revisit it after you apply the activity.

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ACADEMIC READING · SOURCE RECORD

Foundations of Information Systems

Provider	OpenStax
Kind	Peer-reviewed open textbook
Licence / access	CC BY
Official URL	Open official source at openstax.org

Use it this week: Read selectively for this week's foundations and workflow work.

UNIVERSITY OR PROFESSOR VIDEO · SOURCE RECORD

Introduction to JavaScript

Provider	MIT web.lab
Kind	University lecture or official educational video
Licence / access	Access terms apply; video remains the property of its provider and platform.
Official URL	Open official source at youtube.com

Use it this week: Watch with the Data Analysis with Spreadsheets and Python learning outcome in mind; take notes on an idea to test locally.

Turn study into skill

PRACTICE LABORATORY

Complete a guided foundations and digital workflow activity using a Liberian or regional case study.

Keep your work simple, local, and real. A basic note, recording, checklist, screenshot, calculation, or plan is valid practice.

EVIDENCE CHECKLIST

- ☐ I named the main idea.
- ☐ I noted one example.
- ☐ I tried one small action.
- ☐ I saved analysis notebook and dashboard.

SELF-CHECK · BEFORE YOU FINISH

1. Can I explain this week's idea in my own words?
2. What example from Liberia, my community, or a regional service could I connect to it?
3. What is one next action I can test or improve?

REFLECT AND CONNECT

In 3–5 sentences, explain how this week's concept could improve a real organisation, community service, workplace, or learner project. Identify one assumption, one risk, and one action you would test next.

Vocabulary + notebook

Use this final page as a learning checkpoint. Clear terms and brief notes make independent study easier to continue.

SMALL SELF-STUDY GLOSSARY

self-directed study (self-study) means you manage your own learning plan. **learning evidence (evidence)** is the proof you save. A **learning source (source)** is credited to its original provider.

Glossary reference list

evidence – learning evidence: A saved note, plan, calculation, recording, checklist, or other item that shows what you can do.

self-study – self-directed study: A learner plans, studies, practises, and checks understanding without waiting for a live class.

source – learning source: An external reading or video credited to its original provider.

MY NOTE BANK

The one idea I want to remember:

The example I will look for this week:

FINISH WELL

Choose a time for your next study session. Small regular sessions are more useful than waiting for a perfect long study day.

SHARE SAFELY

Discuss an idea with a trusted peer, mentor, or colleague. Keep your own notes and do not share sensitive personal or workplace data.

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